

**Project Title: Consolidating Diffusion Guidance via On-Policy Self-Distillation for Continual Robot Learning****Team Members:** William Liu, Sakthivel Sivaraman, Jerry Gu**Emails:** wzliu@stanford.edu, sakthiv@stanford.edu, jerrygu@stanford.edu

## 1 Objective

Diffusion policies have become a leading approach for visuomotor robot learning, but adapting them to user-desired behaviors at deployment requires either retraining or expensive inference-time intervention. DynaGuide demonstrates that a separately trained dynamics model can steer a frozen diffusion policy toward arbitrary outcome-defined goals via gradient-based guidance during action denoising. However, every successful steering episode is discarded once the rollout completes, and the base policy never accumulates the implicit skill being demonstrated. This statelessness is the limitation we address.

We will investigate whether the implicit knowledge induced by repeatedly running DynaGuide can be consolidated into the base diffusion policy itself through on-policy self-distillation, transforming DynaGuide from a per-rollout intervention into a mechanism for cumulative skill acquisition. Beyond DynaGuide specifically, the question generalizes: when a robot policy is deployed with any inference-time steering signal, can the steering be absorbed into the policy across episodes without per-task adapters or replay storage, while preserving capability on previously-learned tasks?

## 2 Related Work

This project sits between two recent papers.

DynaGuide Du and Song (2025) is the method we will extend. It steers a pretrained diffusion policy at inference time without retraining. A separately trained latent dynamics model predicts future observations in DinoV2 Oquab et al. (2024) embedding space given the current observation and candidate actions, and the gradient of a log-sum-exp distance between the predicted future and a set of guidance condition images modifies the noise prediction inside the action denoising loop. The key idea is decoupling the policy from the guidance signal. Goal-conditioned policies bake the conditioning into training, so any unfamiliar goal pushes the policy out of distribution. DynaGuide stays robust because the encoder, the dynamics model, and the base policy are each individually in-distribution even when the combination faces a new objective. The paper shows this with a 5.4x advantage over goal-conditioning on underspecified objectives.

The result that matters most for us is the underrepresented behavior experiment. With only 1% of training trajectories showing a target behavior, DynaGuide steers to that behavior 40% of the time, while sampling baselines collapse to base policy rates. Active guidance during denoising can find rare modes that sampling cannot surface, since sampling depends on the base policy producing the right action somewhere in its top candidates. The limitation the authors flag is that none of this knowledge is retained. Each rollout is a fresh inference-time intervention, and the base policy never improves from one steering episode to the next. That is the gap we are trying to close.

SDFT Shenfeld et al. (2026) provides the mechanism we use to close it. The setup is on-policy distillation from demonstrations. A context-conditioned version of the model acts as the teacher, the unconditioned model acts as the student, the student samples trajectories on-policy, and the objective is the reverse KL between them. Section 3.1 of the paper shows that this is equivalent to maximizing an implicit IRL objective with reward  $\log \pi(y|x, c) - \log \pi_k(y|x)$ , which has a clear interpretation as a signal to imitate the version of the model that has access to the additional context.

Two findings from the paper shape how we plan to apply it. First, on-policy rollouts are the dominant source of gains. Figure 6 shows that offline distillation from the same teacher consistently underperforms, so the student-sampling step is essential rather than incidental. Second, the teacher must be an EMA of the student. A frozen base teacher stops being useful once the student passes it, and using the current student directly causes training to diverge through feedback amplification. We inherit both choices.

### 3 Technical Outline

**Approach.** We begin with a frozen base diffusion policy Chi et al. (2023) and a frozen DynaGuide dynamics model, both pretrained on CALVIN. During training, the student generates rollouts in CALVIN under no guidance. At every state visited and every denoising step, we record what the teacher’s guided noise prediction would have been, and train the student to produce that prediction unaided. The loss is the per-step squared error between the student’s unguided prediction and the teacher’s guided prediction, with the teacher frozen. Because training data comes from the student’s own rollouts rather than pre-recorded trajectories, the procedure is on-policy in the sense of SDFT Shenfeld et al. (2026), which is what addresses the distribution shift between training-time and deployment-time states. After distillation converges on one task, we proceed to the next task with a new guidance condition, without resetting weights or replaying old data.

**Novel technical contribution.** To our knowledge, this is the first port of SDFT-style on-policy self-distillation from language models producing discrete token output to robot policies that produce continuous actions through diffusion. Additionally, no prior work addresses DynaGuide’s retention limitation directly. The reduction from SDFT’s training objective to a noise-matching loss in the diffusion setting, is part of the contribution.

**Evaluation plan.** We first reproduce DynaGuide on the CALVIN switch\_on task, targeting the 65–75% success rate from the original paper to confirm our setup matches theirs. We then evaluate the distilled policy in three phases: (i) single-task success of the distilled policy without guidance, compared to DynaGuide-with-guidance and the unmodified base policy; (ii) a sequential two-task forgetting analysis, for instance, switch\_on followed by drawer\_open, measuring how much switch\_on success is retained after drawer\_open training; (iii) inference-efficiency comparison between the distilled policy and DynaGuide’s stochastic sampling. Other baselines include supervised fine-tuning on guided rollouts (the off-policy comparison).

### 4 Team Contributions

- **William Liu:** Environment setup and codebase integration; Simulator setup; rollout-collection pipeline coupling CALVIN environment; primary ablations; Transferability to other tasks;
- **Sakthivel Sivaraman:** distillation training loop implementation; Continual-learning scheduler; Catastrophic forgetting analysis;
- **Jerry Gu:** DynaGuide reproduction on switch\_on to verify  $\geq 65\%$  success; SFT-on-guided-rollouts baseline; inference-efficiency comparison;

### References

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